

Investigating Science — Unit Overview

NSW Stage 6

2025-08-26

1 Overview

This unit skeleton is aligned to the NSW Stage 6 Investigating Science syllabus (NESA, 2017). It provides:

- Outcomes table (mapped from syllabus codes)
- Lesson structure placeholders (SI units, British English)
- Assessment map placeholders

References are formatted in APA 7th; specific URLs and DOIs are marked TO-VERIFY where appropriate.

2 Learning Outcomes

The following outcomes (Stage 6) are extracted from the local `learning-outcomes.md` and grouped under Working Scientifically strands. Codes and some placements are TO-VERIFY against the official syllabus PDF.

Code	Strand	Statement
INS11-12-1	Questioning and predicting	Develops and evaluates questions and hypotheses for scientific investigation

Code	Strand	Statement
INS11-12-2	Planning investigations	Designs and evaluates investigations to obtain primary and secondary data and information
INS12-12	Planning investigations; Communicating	Develops and evaluates the process of undertaking scientific investigations
INS11-12-3	Conducting investigations	Conducts investigations to collect valid and reliable primary and secondary data and information
INS11-8	Conducting investigations	Identifies that the collection of primary and secondary data initiates scientific investigations
INS11-12-4	Processing data and information	Selects and processes appropriate qualitative and quantitative data and information using a range of appropriate media

Code	Strand	Statement
INS11-10	Processing data and information	Develops, and engages with, modelling as an aid in predicting and simplifying scientific objects and processes
INS11-12-5	Analysing data and information	Analyses and evaluates primary and secondary data and information
INS11-12-6	Problem solving	Solves scientific problems using primary and secondary data, critical thinking skills and scientific processes
INS11-9	Problem solving	Examines the use of inferences and generalisations in scientific investigations
INS12-14	Problem solving	Uses evidence-based analysis in a scientific investigation to support or refute a hypothesis

Code	Strand	Statement
INS11-12-7	Communicating	Communicates scientific understanding using suitable language and terminology for a specific audience or purpose
INS12-13	Communicating	Describes and explains how science drives the development of technologies
INS11-11	Communicating	Describes and assesses how scientific explanations, laws and theories have developed
INS12-15	Communicating	Evaluates the implications of ethical, social, economic and political influences on science

Note: Strand placements and codes are TO-VERIFY against the current NESA syllabus publication.

3 Lesson Structure (skeleton)

1. Engage: real-world context; clarify prior knowledge; elicit questions.
2. Explore: design or critique an investigation plan; identify variables; ensure SI units.
3. Explain: analyse sample data; practise modelling and error analysis.
4. Elaborate: extend to unfamiliar contexts; connect to technology and society.

5. Evaluate: formative checks; map to outcomes; literacy/numeracy progression notes.

4 Assessment Map (skeleton)

- Task A (Depth Study brief) — maps to INS11-12-2, INS11-12-3, INS11-12-5, INS11-12-7. TO-VERIFY weightings.
- Task B (Data analysis) — maps to INS11-12-4, INS11-12-6, INS12-14. TO-VERIFY weightings.
- Exam-style items — link to past HSC papers (2019–2024) for practice; question-to-outcome mappings are TO-VERIFY.

5 References

See bibliography entries (APA 7th) in `references.bib`.